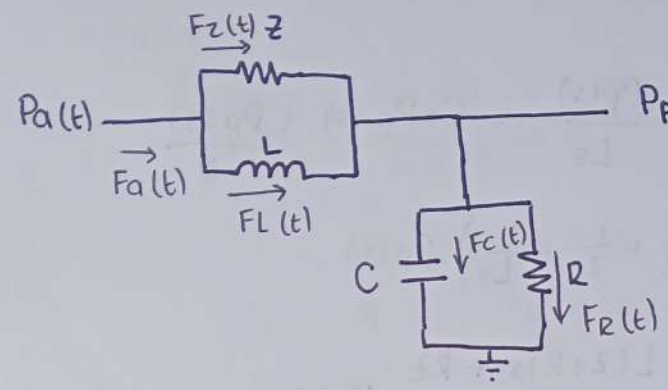
Práctica 5.4 - Modelado y Control del Sistema Cardiovasuular usando el modelo Windkessel



Modelo Windkessel

Función de Transferencia

$$Fa(t) = Fz(t) + FL(t) = Fc(t) + FR(t)$$

$$Fz(t) = \frac{Pa(t) - Pp(t)}{Z}$$

$$FL(t) = \frac{1}{L}\int [Pa(t) - Pp(t)]\, dt$$

$$Fc(t) = C\,\frac{dPp(t)}{dt}$$

$$FR(t) = \frac{Pp(t)}{R}$$

Transformada de Laplace

$$Fa(s) = Fz(s) + FL(s) = Fc(s) + FR(s)$$

$$Fz(s) = \frac{Pa(s) - Pp(s)}{Z}$$

$$FL(s) = \frac{Pa(s) - Pp(s)}{Ls}$$

$$Fc(s) = CsPp(s)$$

$$FR(s) = \frac{Pp(s)}{R}$$

$$\frac{Pa(s) - Pp(s)}{Z} + \frac{Pa(s) - Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

Análisis Algebraico

$$\frac{Pa(s)}{Z} - \frac{Pp(s)}{Z} + \frac{Pa(s)}{Ls} - \frac{Pp(s)}{Ls} = CsPp(s) + \frac{Pp(s)}{R}$$

$$\left(\frac{1}{Z} - \frac{1}{Ls}\right) Pa(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right) Pp(s)$$

$$\frac{Ls+Z}{LZs} Pa(s) = \frac{CLRZs^2 + L(Z+R)s + RZ}{LRZs} Pp(s)$$

$$\frac{Pp(s)}{Pa(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{LRZs}}$$

$$\frac{Pp(s)}{Pa(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$

Error en estado estacionario

Este error se calcula cuando la entrada $Pa(s)$ es un escalón unitario, es decir, $Pa(s) = 1/s$

$$e(s) = \lim_{s \to 0} sPa(s)\left(1 - \frac{Pp(s)}{Pa(s)}\right) = \lim_{s \to 0} s\frac{1}{s}\left(1 - \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}\right) = 0$$

$$e(t) = 0v$$

Estabilidad del sistema en lazo abierto

Se deben calcular los polos, es decir, las raíces del denominador.

$$CLRZs^2 + L(Z+R)s + RZ = 0$$

$$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

$a = CLRZ$

$b = L(Z+R)$

$C = RZ$

Caso Hipotenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 3.500$

$$\frac{Pp(s)}{Pa(s)} = \frac{0.001s + 0.010}{3.5 \times 10^{-5} s^2 + 0.00101s + 0.01}$$

$$\lambda_1 = -14.429 + 8.805\ i$$
$$\lambda_2 = -14.429 - 8.805\ i$$

Respuesta Subamortiguada.

Caso Normotenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.500$

$$\frac{Pp(s)}{Pa(s)} = \frac{0.001s + 0.01}{1.5 \times 10^{-5} s^2 + 0.00101s + 0.01}$$

$$\lambda_1 = -12.062$$
$$\lambda_2 = -55.272$$

Respuesta estable sobreamortiguada

Caso Hipertenso

$Z = 0.010$
$R = 1.000$
$L = 0.001$
$Z = 1$

$$\frac{Pp(s)}{Pa(s)} = \frac{0.001s + 0.01}{0.00001s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -10 \qquad \lambda_2 = -10$

Respuesta estable críticamente amortiguada.

Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]\,dt = C\frac{d\,Pp(t)}{dt} + \frac{Pp(t)}{R}$$

Nota: la unica variable desconocida es la salida del sistema $Pp(t)$. NO se puede despejar de términos en los que se esté derivando o integrando.

$$\frac{Pa(t)}{Z} + \frac{1}{L}\int\left[Pa(t) - Pp(t)\right]dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{R}$$

$$Pa(t) + \frac{1}{L}\int\left[Pa(t) - Pp(t)\right]dt - C\frac{dPp(t)}{dt} = \frac{Pp(t)}{R} + \frac{Pp(t)}{Z}$$

$$Pp(t) = \left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int\left[Pa(t) - Pp(t)\right]dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t)\ \frac{Z+R}{RZ} = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int\left[Pa(t) - Pp(t)\right]dt - C\frac{dPp(t)}{dt}\right]$$

$$Pp(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int\left[Pa(t) - Pp(t)\right]dt - C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$